

EEE - 4484

Student ID: 220041125

Name : Nageemul Hasan Prince

Section : 1A

Expt Name : Study of different logic family

Expt No : 01

①

Objective:

- i) Learn and understand internal circuit layouts of several logic Families, with an emphasis on Diode logic (DL) and Diode-Transistor Logic (DTL)
- ii) Examine the operation of a transistor based NOT gate
- iii) Derive truth table to create a combinational logic function at gate level using AND, OR, NOT gates,

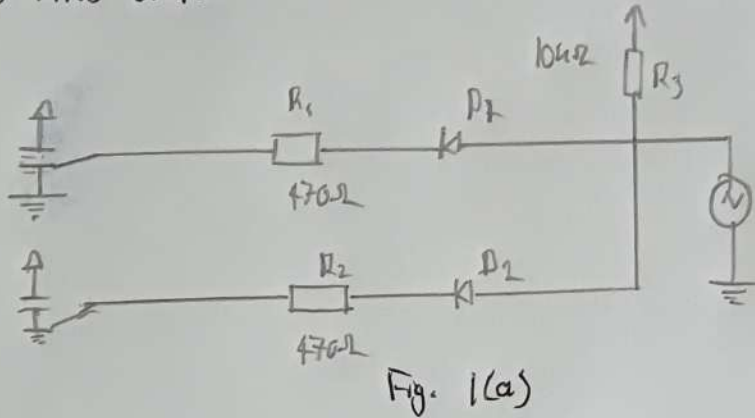
Requirements:

1. DC power supply
2. Bread board
3. Resistors
4. Diodes
5. Transistors (BJT)
6. Multimeter
7. Connecting wires

②

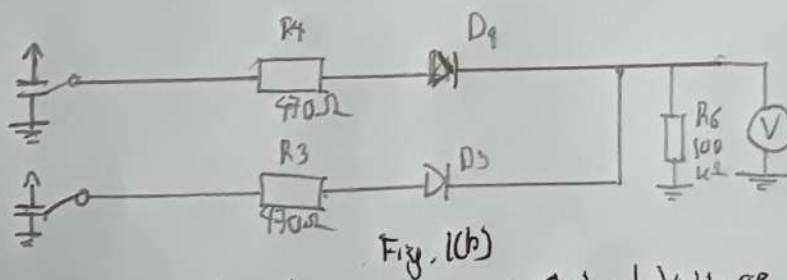
Circuit Diagram and Truth Table

i) AND Gate:



Input 1	Input 2	Output Voltage	Logic Output Level
0	0	0.60V	0
0	1	0.730	0
1	0	0.739	0
1	1	5.04 5.04	1

ii) OR Gate:



Input 1	Input 2	Output Voltage	Logic Output Level
0	0	0	0
0	1	4.5 3.91	1
1	0	4.8 3.94	1
1	1	4.2	1

③

NAND Gate:

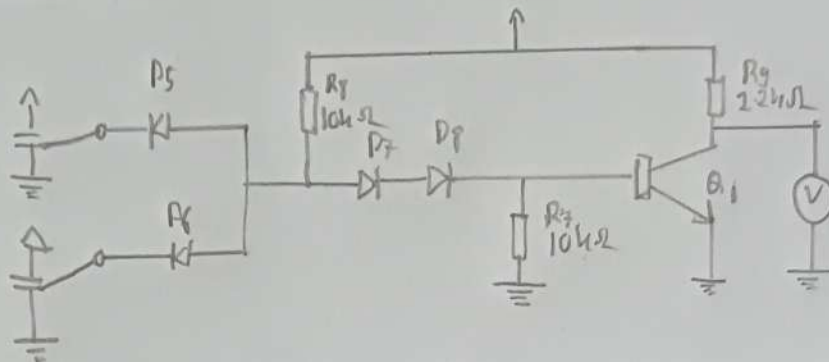


Fig. 1(c)

Input 1	Input 2	Output Voltage	Logic Output Level
0	0	5.05	1
0	1	5.05	1
1	0	5.05	1
1	1	0.05	0

NOR Gate:

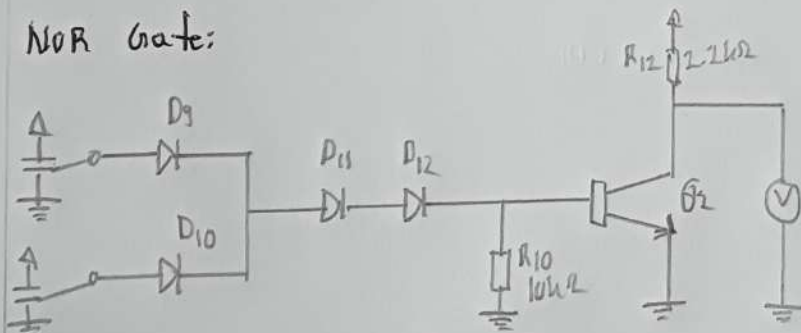


Fig. 1(d)

Input 1	Input 2	Output Voltage	Logic Output Level
0	0	5.05	1
0	1	0.01	0
1	0	0.01	0
1	1	0.01	0

④

NOT Gate:

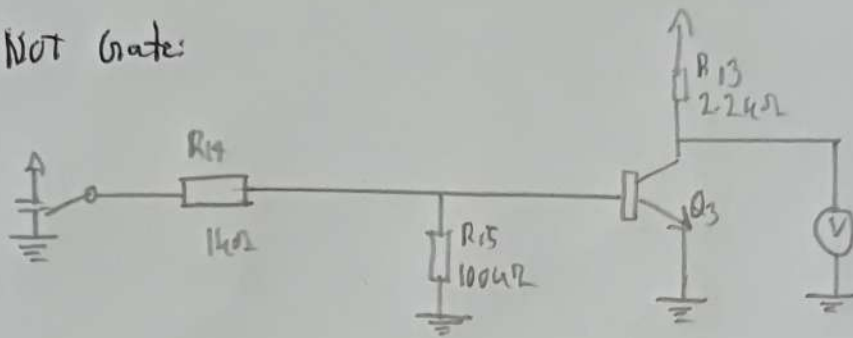


Fig. 1(e)

Input	Output Voltage	Logic Output Level
0	5.05	1
1	0.01	0

⑤

Justifications:

1. AND Gate: (Diode Logic)

An AND gate in DL is made with diode such that the output is high only if every input is high. The matching diodes conduct and pull output low if any input is low.

Result:

- i) Output is high only when all inputs are high.
- ii) Otherwise, output is low.

2. OR Gate: (Diode Logic)

Arrangement of diode in OR gate permits current to pass in the event that only any input is high - resulting in a high output. Output is low if all inputs are low.

Result:

- i) Output is high when at least one input is high
- ii) Output is low when all inputs are low.

(6)

3. NOT Gate:

A transistor is used to design an inverter. The transistor acting as at one state pulls the output down when input is high. Transistor is turned off when input is low - resulting in high output.

Result:

- i) High input results in low output
- ii) Low input results in high output.

4. NAND Gate: (Diode-Transistor Logic)

NAND is a combination of AND and NOT Gate. AND function is handled by diode and the output is inverted by a transistor.

Result:

- i) Output is always high except when both inputs are high.

⑦

5. NOR Gate: (Diode Transistor Logic)

Circuit configuration has a diode acting as OR Gate and a transistor inverts the signal.

Result:

i) Output is always low except when both inputs are high, low,

8

Answers to Report Questions

1. Logic '1' = High Voltage

Logic '0' = Low Voltage

a) AND Gate:

<u>A</u>	<u>B</u>	<u>Output</u>
0	0	0
0	1	0
1	0	0
1	1	1

Output is 1 only when both inputs are 1.

b) OR Gate:

<u>A</u>	<u>B</u>	<u>Output</u>
0	0	0
0	1	1
1	0	1
1	1	1

Output is 1 only when at least one input is 1

c) NOT Gate:

<u>A</u>	<u>Output</u>
0	1
1	0

Output is inversion of input

9

d) NAND Gate:

A	B	Output
0	0	1
0	1	1
1	0	1
1	1	0

Output is 1 unless both inputs are 1

e) NOR Gate:

A	B	Output
0	0	1
0	1	0
1	0	0
1	1	0

Output is 1 only when both inputs are 0

2. Fig. 1(d) is the depiction of NOR Gate. NOR is the combination of OR and NOT Gate. The OR Function is performed by the two diodes. As we give two inputs, each diode catches one input each. If either input is 1, the diode conducts.

In short, we need two diodes to cover all paths as we are giving two inputs.

(10)

3. The general forward voltage threshold for common silicon diodes are roughly $0.7V$.

Fig. 1(a) (AND), Fig. 1(b) (OR), Fig. 1(c) (NAND), Fig. 1(d) (NOR)

all uses basic ~~st~~ silicon diodes as logic elements.

Thus for all these, the minimum voltage required for forward bias is $0.7V$.

4. A NOT Gate is constructed with BJT in common emitter mode [Fig. 1(e)]

• When input is low:

Collector doesn't conduct since transistor is off. A linked resistor pulls the output node High. Thus low input results in high output.

• When input is high:

After turning on, resistors reach saturation. It pulls the output low by conducting extensively and establish channel to ground. Thus output is low when input is high.

(11)

Discussion:

Using diodes and transistors, we learned and executed AND, OR, NAND, NOR and NOT Gates. These experiments aim at DL and DTL Families. Furthermore, we practically reviewed and compared results with logic table,

Additionally, we saw that a minimum threshold of 0.7V was required to conduct. We also noticed that logical high was not always accurately 5V or low was not accurately 0V.

In whole, this lab gives an overview how logic operations are executed underneath and focused on digital logic design.